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Gaining Popularity through Sponsorship in the Open Source Software Community

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Gaining Popularity through Sponsorship in the Open Source Software Community

Completed Research Paper

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Abstract

Peer-to-peer sponsorship—the monetary donation from one user to another—is a growing trend on online community platforms. However, existing research primarily focuses on the sponsee perspective. The potential gains for sponsors providing sponsorship remain largely unknown. Using a large panel dataset collected over 52 months from GitHub, the world’s largest open source software (OSS) community, we examine the impact of sponsoring others on the popularity of an individual’s OSS projects and how the individual’s tenure and disclosure of affiliations within the community moderate this relationship. Our findings reveal a positive impact of providing sponsorship on project popularity. Moreover, individuals with shorter tenure and those who do not publicly disclose affiliations experience a greater increase in project popularity following their sponsorship. Our study is among the first to examine the gains for sponsors in peer-to-peer sponsorship, offering valuable insights for online communities and their members.

Keywords: open source software community, sponsorship, sponsor, project popularity

Introduction

In recent years, an increasing number of online community platforms have incorporated the peer-to-peer sponsorship feature to incentivize user engagement (Yoon et al., 2022). For instance, Zhihu, a prominent Chinese question-and-answer platform, introduced Zhihu Live in 2016, facilitating users to host live streams and receive sponsorship from their audience. Similarly, GitHub, the world's largest open source software (OSS) community, initiated GitHub Sponsors in 2019, allowing users to sponsor each other financially (Zuegel, 2019). This novel feature fosters the development of more sustainable and self-sufficient online communities by providing a mechanism through which users can support each other financially.

The advantages of peer-to-peer sponsorship for sponsees (i.e., users receiving sponsorship) and the community platform are apparent. However, the critical question is how to incentivize users to sponsor others. It is crucial to establish that sponsors do not merely provide support but can also reap advantages, such as heightened popularity of the content or projects they create, from their sponsorship. On social media platforms like Facebook, users actively pursue likes on their posts, which act as recognition from other users (Lin & Lu, 2011). Similarly, users in OSS communities like GitHub value the stars awarded to their OSS projects (i.e., public repositories), as stars symbolize recognition from the community for their OSS work. These stars serve as a gauge of project popularity (Weng & Soh, 2023), playing a crucial role in determining a project's ranking on the platform. By demonstrating the potential for sponsors to attain heightened project popularity through their sponsorship, our study may incentivize users who prioritize peer recognition for their content or projects to initiate or sustain sponsorship activities.

Despite the critical role that sponsors play in peer-to-peer sponsorship, existing studies on online peer-to-peer sponsorship have predominantly focused on the sponsee side, such as the determinants of sponsorship acceptance and the impacts of sponsorship on sponsees (Krishnamurthy et al., 2014; Medappa et al., 2023; J. Wang et al., 2022; Yoon et al., 2022). Research on the sponsor side has only examined the determinants of sponsorship decisions (Krishnamurthy & Tripathi, 2009; Shimada et al., 2022) and the impacts of sponsorship on sponsors' perception and behavior (Y. Wang et al., 2022; Yoon et al., 2022), while little is known regarding the potential benefits sponsors may derive from providing sponsorship. Our study aims to address this research gap by shedding light on the potential gains for sponsors. Specifically, we study whether sponsoring other developers in the OSS community enhances an individual's project popularity, as well as the moderating factors in this relationship, based on the data from GitHub. This research topic sheds light on the potential benefits of sponsoring others and helps to identify the individuals who can gain the most from the sponsoring activities. This leads us to our first research question: *Does sponsoring other users affect the popularity of an individual's OSS projects?*

The answer to this question is not clear. On the one hand, the act of sponsoring may not necessarily influence the popularity of sponsors' projects for the following reasons. First, community members may perceive sponsorship as a private transaction between two individuals, leading to a lack of incentive to star projects belonging to sponsors who are not directly related to them. Second, OSS project popularity is primarily shaped by the quality and usefulness of the project itself (Dabbish et al., 2012). Thus, the act of sponsoring other developers does not inherently prompt users to star the sponsor's OSS projects.

On the other hand, according to the signaling theory (Connelly et al., 2011; Spence, 1978), the act of sponsoring functions as a signal that conveys a positive impression of the sponsor – someone who is actively engaged, generous, and committed to open source. This positive image, once established through sponsorship, encourages community members to explore the sponsor's projects more attentively. Community members may evaluate these projects through the lens of the sponsor's positive attributes. They may also choose to star the sponsor's projects as an expression of appreciation and recognition toward the sponsor's supportive behavior, given the importance of stars in the OSS community (Dabbish et al., 2012; Weng & Soh, 2023). Hence, we propose that sponsoring others could enhance project popularity.

Furthermore, some individual characteristics could moderate the main impact of providing sponsorship on project popularity. Therefore, we ask our second research question: *What factors moderate the impact of providing sponsorship on project popularity?* We examine whether an individual's tenure in the community and affiliation disclosure influence the signaling effect of providing sponsorship on the popularity of the individual's projects. First, individuals with shorter tenure may benefit more from sponsoring others due to the signaling effect. Newcomers with shorter tenure may use sponsorship as a way to quickly establish their reputation and gain recognition within the community. In contrast, individuals

with longer tenure may already have established themselves as committed members of the community, so the impact of their sponsorship actions on project popularity may be less pronounced. Second, individuals who do not disclose their company affiliation may be perceived as more dedicated to open source, as their sponsorship actions are seen as genuine expressions of intrinsic motivation rather than driven by corporate interests. This perception may enhance the positive impression of their sponsorship behavior, leading to increased project popularity.

An underlying challenge in identifying these effects is the possible endogeneity issues in drawing causal inference regarding the impact of providing sponsorship. While an ideal approach may involve a randomized experiment where the right to sponsor others is randomly assigned to all platform users, this is not an ideal identification strategy in our context. One major reason is that our objective is to examine the causal changes experienced by naturally occurring sponsors. Additionally, conducting such a randomized experiment may be unethical and not in the best interest of developers who could have been sponsored but were not, as well as the platform itself (Jung et al., 2019). Hence, we are limited to working with observational data and employ multiple causal identification strategies, including staggered difference-in-differences (DID) and propensity score matching (PSM), to address this challenge. Through various empirical analyses, we consistently observe a positive effect of sponsoring others on project popularity. We also find that this effect is larger for individuals with shorter tenure in the community and those without announced affiliations.

Our study offers several contributions to the existing literature. First, drawing on the signaling theory (Connelly et al., 2011; Spence, 1978), our study provides evidence that sponsoring behavior serves as a powerful signal of generosity and commitment, ultimately resulting in an amplification of project popularity. Furthermore, our study offers a nuanced understanding of the moderating role of individuals' tenure and affiliation disclosure on the effectiveness of these signals. Second, our work expands the literature on online peer-to-peer sponsorship by examining the benefits that sponsors can gain from sponsoring others. While previous research has predominantly focused on the sponsee side (Krishnamurthy et al., 2014; J. Wang et al., 2022), our study sheds light on the potential advantages that sponsors may gain through providing sponsorship. As such, our study offers a unique and insightful contribution to the field and furthers our understanding of the outcomes of engaging in online peer-to-peer sponsorship activities. Third, our study contributes to the OSS literature by being among the first to explore the causal effects of offering peer-to-peer sponsorship on sponsors' gains in the OSS community. Given the relatively recent emergence of this phenomenon, there has been limited exploration of its impacts, especially on sponsors.

Our study also has practical implications for both online communities and their users. First, our findings suggest that introducing a peer-to-peer sponsorship feature could be a valuable strategy for community platforms. Such a feature not only better motivates sponsees to actively participate and contribute but also enhances the popularity of sponsors' projects. Platforms can leverage our study to facilitate user sponsorship, creating a more vibrant community environment. Second, our study reveals that individuals with shorter tenure and without announced affiliations experience a more pronounced surge in project popularity when engaging in sponsorship activities. Platforms can utilize these insights to encourage new members and individuals who prefer not to disclose affiliations to sponsor others, as it could help them to gain recognition more quickly. Third, our study provides insights to community members that sponsoring others can serve as an effective strategy to increase their visibility and project popularity, particularly for newcomers and those without announced affiliations. Our findings guide community members in making informed decisions regarding whether to sponsor others and help them understand the potential benefits of doing so.

Literature Review

In this section, we review the online peer-to-peer sponsorship literature from both the sponsee and sponsor sides. Figure 1 shows the positioning of our research. Prior literature has mainly studied from the sponsee side. Research on the sponsor's side is limited to examining the factors that determine sponsorship decisions and the effects of sponsorship on sponsors' perception and behavior. However, it is unknown whether sponsors can gain from others after providing sponsorship.

Prior literature has suggested that many content or project creators turn to sponsorship as a means of generating income and exposure (Gerber & Hui, 2013), as they believe that peer-to-peer sponsorship can

provide them with more creative freedom and autonomy. In the context of OSS communities, there is increasing evidence showing that OSS contributors wish to get sponsorship for their open source work (Krishnamurthy et al., 2014; Zuegel, 2019). Sponsorship can provide sponsored developers with access to new tools and technologies that they may not have been able to afford otherwise, which can help them improve their skills and contribute more effectively to OSS projects (Lakhani & Wolf, 2005). Krishnamurthy et al. (2014) have found that intrinsic factors and extrinsic motivations positively influence developers' willingness to accept financial sponsorship, while community motivation negatively influences their willingness to accept monetary rewards.

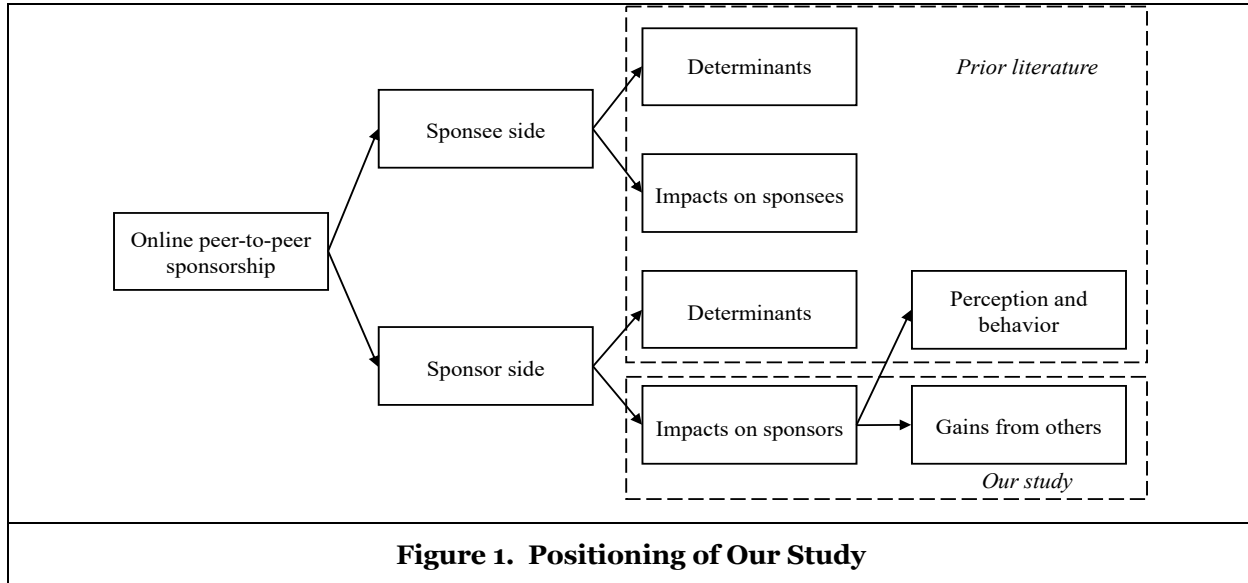


Figure 1. Positioning of Our Study

The impacts of online peer-to-peer sponsorship on sponsees have been extensively studied in the literature. Shimada et al. (2022) surveyed 177 sponsored developers on GitHub and found that they were more active than those who had not received sponsorship. Y. Wang et al. (2022) found that after receiving sponsorship on GitHub, the sponsees' projects become more productive and healthier with fewer bugs reported. Moreover, Medappa et al. (2023) showed that being sponsored on GitHub increases contributions to maintenance-related tasks. In addition to the effects on OSS developers, the impact of peer-to-peer sponsorship on sponsees on online knowledge-sharing platforms has also been studied (Kuang et al., 2019; J. Wang et al., 2022). Kuang et al. (2019) found that Zhihu users who receive sponsorship from other users are more likely to voluntarily share knowledge on the platform. Similarly, J. Wang et al. (2022) showed that receiving sponsorship increases the quantity and quality of knowledge spillover for Zhihu Live hosts. Our study differs from this stream of research by focusing on the impact of online peer-to-peer sponsorship on sponsors – the other side of sponsorship.

Research from the sponsor side has mainly focused on the determinants of their sponsorship decisions. Shimada et al. (2022) have identified determinants such as the perceived value of the project or developer, the sponsor's personal interest, and the sponsor's financial situation. Zhang et al. (2022) showed that the most significant factor influencing the amount of sponsorship is the social status of the sponsored developer. Eghbal (2020) suggested that sponsors on GitHub are not motivated by altruism but by their interest in the sponsee's future work based on their current reputation. Prior research has also shown that sponsors are motivated by interpersonal connections between the sponsor and the sponsee (Gerber et al., 2012) as well as personal needs such as recognition, belonging, and making a positive impact on society (Gerber & Hui, 2013).

However, only a few studies have investigated the impact of peer-to-peer sponsorship on sponsors. Research has shown that engaging in prosocial behavior, such as sponsoring others, can lead to greater happiness and well-being for the sponsors themselves (Aknin et al., 2013; Dunn et al., 2008). Besides, sponsorship has been found to influence sponsors' behavior in online communities (Y. Wang et al., 2022; Yoon et al., 2022). Y. Wang et al. (2022) found that sponsors become more active on GitHub after sponsoring others. Yoon et al. (2022) found that users who sponsor a pizza to others on Reddit increase

their engagement behavior by writing more posts and comments compared to non-sponsors. Despite the existing research on the impacts of sponsorship on sponsors' perception and behavior, there is a research gap in understanding whether sponsors gain from others after offering sponsorship. Thus, our study aims to contribute to the literature by investigating the effect of online peer-to-peer sponsorship on sponsors' gains from others, with a focus on project popularity.

Hypothesis Development

According to the signaling theory (Connelly et al., 2011; Spence, 1978), a signal sender can credibly convey some information about itself to a signal receiver by sending signals. In our specific context, GitHub users can credibly convey information that they are actively engaged, generous, and committed to open source by sponsoring OSS developers.

After an individual becomes a sponsor of a developer on GitHub, three distinct changes will appear on his or her profile page – a “sponsor” badge and two “sponsoring” labels. These changes satisfy the two primary conditions for being a signal, namely, information content and observability (Foerderer et al., 2021). First, the act of sponsoring conveys quality-related information. Sponsoring other OSS developers is an outstanding and appreciated behavior, indicating to community members that the sponsor is generous and has contributed to open source in a significant way (Zuegel, 2019). The changes on a user's profile page after they become a sponsor are considered a reliable signal as they are conferred by the platform. The “sponsor” badge and “sponsoring” labels will never appear on non-sponsors' pages, making it clear to others that the user has demonstrated their commitment to supporting the OSS community. Second, the act of sponsoring others is observed by anyone who visits the user's GitHub page. When people click on a user's avatar or name, they are redirected to the user's profile page, where they can easily see if there is a “sponsor” badge or label.

The act of sponsoring operates as a signal, creating a positive impression of the sponsor. This positive image, characterized by qualities such as generosity and commitment, entices community members to delve into the sponsor's projects with heightened interest and attention. This engagement stems from an intrinsic curiosity spurred by the perceived virtues associated with the sponsor. When community members encounter the sponsor's projects, they might evaluate these projects not only for their inherent usefulness but also through the lens of the sponsor's positive attributes (Weng & Soh, 2023). Community members may also choose to bestow stars as a heartfelt gesture of appreciation and recognition toward the sponsor's benevolent actions (Krishnamurthy & Tripathi, 2009). This resonates with the psychological principle of reciprocity, where the community responds positively to the sponsor's constructive behavior with positive responses. Accordingly, we propose our first hypothesis:

H1: Providing sponsorship has a positive impact on project popularity.

Next, we will delve deeper into the boundary condition of the signaling effect of sponsoring others. The effect of sponsoring others on project popularity may be larger for individuals with shorter tenure due to several reasons. First, individuals' tenure in the community may reduce the fit between the information conveyed by the sponsoring behavior and the extent to which community members can observe what is being signaled (Ho & Rai, 2017). Since signal fit positively affects signal effectiveness (Connelly et al., 2011), the effectiveness of the signal associated with the sponsor's commitment to open source decreases with tenure. As individuals with longer tenure have already built a reputation for commitment to open source, it is natural for them to sponsor OSS developers. In contrast, it may come as a surprise to community members to see individuals with shorter tenure sponsoring others. Therefore, the signaling effect of sponsoring others is larger for sponsors with shorter tenure.

Second, individuals with longer tenure are usually more committed and loyal members of the OSS community (Krishnamurthy et al., 2014; Krishnamurthy & Tripathi, 2009; Stewart & Gosain, 2006). They have invested significant time and effort in contributing to the community and may have a strong sense of belonging within it. This sense of belonging may reduce their reliance on external validation, such as project popularity. Therefore, community members may believe that sponsors with longer tenure will continue to contribute to open source, even if they do not explicitly star those sponsors' projects. In contrast, community members may think it is more necessary to show recognition to sponsors with shorter tenure as they need more feedback and recognition to maintain their motivation to contribute to the community (Burke et al., 2009). Therefore, we propose an alternative hypothesis:

H2: The effect of providing sponsorship on project popularity is larger for individuals with shorter tenure in the community.

Our second moderator involves the disclosure of one’s affiliated company. Individuals who choose not to announce affiliations demonstrate an elevated commitment to open source. Their sponsorship actions take on a heightened signaling role, where they signal not only their personal dedication but also their independence from external associations.

In online communities, users may perceive those affiliated with certain companies to receive preferential treatment or biased attention due to their corporate associations. Community members may attribute the sponsoring actions of individuals affiliated with companies to corporate interests rather than personal dedication, as such actions can increase brand awareness for companies. On the contrary, individuals without announced affiliations are unburdened by such expectations. Their sponsorship initiatives are construed as genuine expressions of intrinsic motivation (Shah, 2006), intended to nurture OSS development and support fellow developers. By effectively mitigating any potential biases stemming from company associations (Dahlander & Wallin, 2006; Zhou et al., 2016), these sponsors enjoy a reputation of impartiality and objectivity. This reputation, in turn, serves to amplify the positive impression conveyed through their sponsorship behavior. This heightened positive image consequently prompts community members to engage more profoundly with the sponsors’ projects and star them. Accordingly, we propose the following hypothesis:

H3: The effect of providing sponsorship on project popularity is larger for individuals without announced affiliations.

Data

We collected data from GitHub, which is the world’s largest OSS community. Our dataset takes advantage of a unique feature of GitHub called GitHub Sponsors, which was introduced on May 23, 2019 and enables users to provide financial support to developers (Zuegel, 2019). Each user on GitHub can become a sponsor for fellow developers by selecting from various sponsorship tiers that are available on the sponsored developer’s “sponsors page”. Upon providing financial support to other GitHub users, sponsors will automatically receive a “sponsor” badge on their profile pages. Furthermore, the “sponsoring” labels will appear in two distinct locations on their profile pages, indicating the developers they are sponsoring. A sample sponsor’s profile page is presented in Figure 2, with all changes emphasized by red boxes.

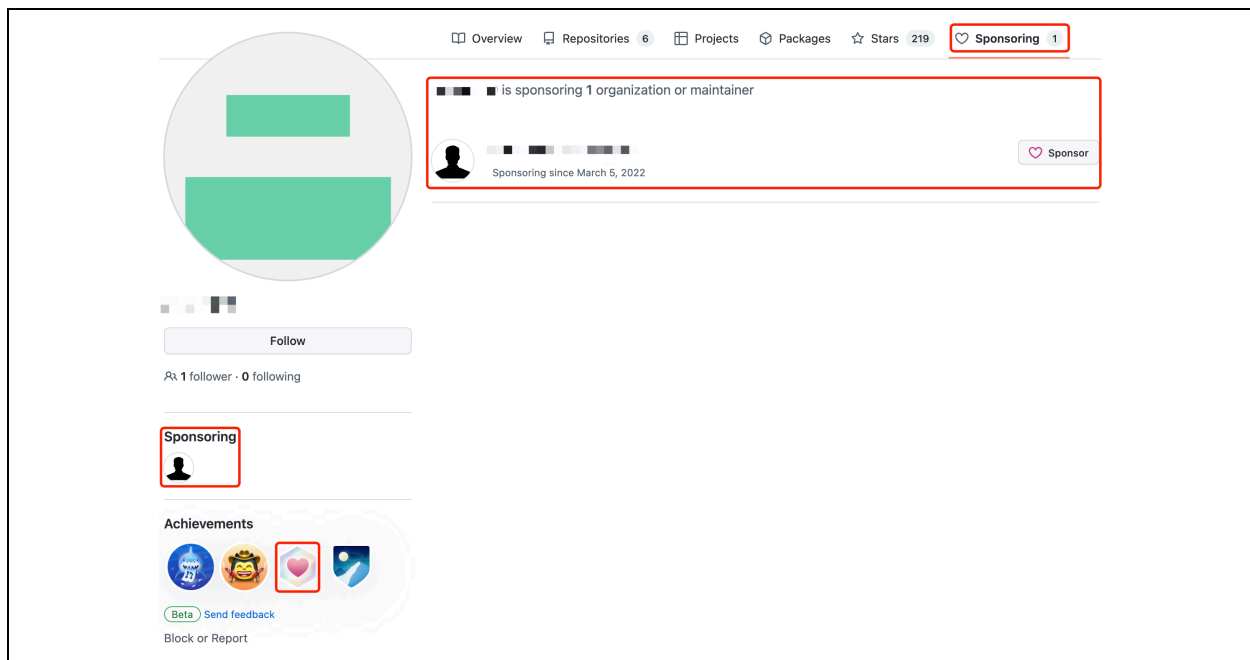


Figure 2. A Screenshot of a Sponsor’s Profile Page on GitHub

Our data was obtained using GitHub API and from GH Archive. We first extracted all the users who were active on GitHub in June 2019, the month just after the introduction of GitHub Sponsors, from GH Archive. We then used the `sponsorshipsAsSponsor` field of the GraphQL API to obtain a list of all sponsorships made by these active users, along with their corresponding sponsees and start dates. From the pool of 2,740,295 active users in June 2019, after excluding users who have cancelled their GitHub accounts, we identified a treatment group of 12,520 sponsors and randomly sampled 100,000 non-sponsors as a control group. After collecting data on the fundamental information and historical activities of all 112,520 sampled users, we created an individual-month level panel that covers the period between May 23, 2018 (one year before the introduction of GitHub Sponsors), and September 22, 2022 (when we started collecting the data).

Empirical Analysis

The Impact of Providing Sponsorship on Project Popularity

In this section, we examine the impact of providing peer-to-peer sponsorship on project popularity using the following staggered DID specification:

$$star_{it} = \beta_0 + \beta_1 sponsor_{it} + \beta_2 controls_{it} + \mu_i + \theta_t + \varepsilon_{it} \quad (1)$$

where the dependent variable, $star_{it}$, is the number of stars that user i receives for his or her OSS projects in month t , and it is a commonly accepted proxy for OSS project popularity (Dabbish et al., 2012; Weng & Soh, 2023); $sponsor_{it}$ is a dummy variable indicating whether user i sponsors someone in month t ; $controls_{it}$ include (i) $tenure_{it}$, the number of months since user i created GitHub account until month t , (ii) $repo_{it}$, the cumulative number of OSS projects created by user i in month t , (iii) $sponsored_{it}$, the number of sponsorships received by user i in month t , (iv) $public_{it}$, the number of user i 's repositories turned from private to public in month t , and (v) $release_{it}$, the number of releases in user i 's OSS projects in month t (Borges & Tulio Valente, 2018; Krishnamurthy & Tripathi, 2009); μ_i is individual fixed effects; θ_t is time fixed effects; and ε_{it} is the error term.

Table 1. Impacts of Providing Sponsorship on Project Popularity			
	Dependent variable: <i>star</i>		
	(1)	(2)	(3)
<i>sponsor</i>	1.089*** (0.407)	5.100*** (1.369)	2.027*** (0.584)
<i>sponsor × tenure</i>		-0.037*** (0.012)	
<i>sponsor × company</i>			-1.778** (0.832)
<i>tenure</i>	-0.041 (0.076)	-0.039 (0.076)	-0.041 (0.076)
<i>project</i>	0.003** (0.001)	0.003** (0.001)	0.003** (0.001)
<i>sponsored</i>	0.088 (0.291)	0.090 (0.291)	0.088 (0.291)
<i>public</i>	2.170*** (0.291)	2.171*** (0.291)	2.170*** (0.291)
<i>release</i>	0.473*** (0.130)	0.473*** (0.130)	0.474*** (0.130)
Constant	2.394 (4.795)	2.305 (4.794)	2.399 (4.796)
Individual FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Observations	5,851,040	5,851,040	5,851,040

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Heteroskedasticity-robust standard errors in parentheses.

The estimation results for Equation (1) are presented in Column (1) of Table 1. We find that the coefficient of $sponsor_{it}$ is significantly positive, supporting H1. Compared to non-sponsors, users who do sponsor someone experience an increased level of project popularity than before, as their projects receive an average of 1.089 more stars than non-sponsors' projects.

It is worth noting that the key argument of the DID method is the parallel trends assumption, which posits that the average change in the control group is assumed to represent the counterfactual change in the treatment group if there had been no treatments. In our specific context, an endogeneity concern is that users are self-selected to provide sponsorship, and therefore, the parallel trends assumption may not hold. Thus, we conduct various analyses to address this endogeneity concern in the following sections.

Moderating Factors

In this section, we first examine how individuals' tenure in the community moderates the impact of providing sponsorship on project popularity using the following specification:

$$star_{it} = \beta_0 + \beta_1 sponsor_{it} + \beta_2 (sponsor_{it} \times tenure_{it}) + \beta_3 controls_{it} + \mu_i + \theta_t + \varepsilon_{it} \quad (2)$$

where $tenure_{it}$ is the number of months since user i joined GitHub until month t . We have the same controls as in Equation (1). The estimation results are presented in Column (2) of Table 1. The coefficient of the interaction term is significantly negative, which implies that individuals with shorter community tenure experience a greater increase in project popularity following their sponsorship, in comparison to those with longer tenure. Thus, H2 is supported.

Then, we examine how the disclosure of individuals' company affiliation moderates the impact of providing sponsorship on project popularity using the following specification:

$$star_{it} = \beta_0 + \beta_1 sponsor_{it} + \beta_2 (sponsor_{it} \times company_i) + \beta_3 controls_{it} + \mu_i + \theta_t + \varepsilon_{it} \quad (3)$$

where $company_i$ is a dummy that indicates whether user i is affiliated with a company as shown on his or her profile page. We have the same controls as in Equation (1). Note that $company_i$ is not included as a standalone variable because μ_i absorbs its effect. The estimation results are presented in Column (3) of Table 1. The significantly negative coefficient of the interaction term suggests that the positive effect of providing sponsorship on project popularity is larger for individuals without announced affiliations than those with an affiliated company. Thus, H3 is supported.

Addressing Endogeneity Concerns

Pre-Treatment Trends

One potential concern in the DID model is the presence of heterogeneity in pre-treatment trends between control and treatment groups. Such heterogeneity could violate the parallel trends assumption. To address this concern, we use the relative time model (Bapna et al., 2018):

$$star_{it} = \beta_0 + \beta_1 T_{nit} + \beta_3 controls_{it} + \mu_i + \theta_t + \varepsilon_{it} \quad (4)$$

where T_{nit} are series of dummies that indicate the chronological distance between the observation and the actual month when user i starts providing sponsorship. Following Bapna et al. (2018), we set the range of n to minus three to three, which represents the periods from three months before to four months after user i starts providing sponsorship. $T + 0$ is a dummy variable for the month when user i starts providing sponsorship. All the other variables are the same as those in Equation (1).

The estimation results for Equation (4) are presented in Table 2. The absence of significance in the $T - 3$, $T - 2$, and $T - 1$ periods indicates that there was homogeneity between the sponsors and non-sponsors prior to the treatment. This suggests that the sponsors did not experience any unusual increase in the number of stars before they sponsor others. However, the positive and significant signs observed in $T + 0$, $T + 1$, and $T + 2$ are consistent with our estimated treatment effect, providing further support for our findings.

Placebo Test

Table 2. Relative Time Model

	Dependent variable: <i>star</i>
<i>T</i> - 3	1.009 (0.620)
<i>T</i> - 2	0.289 (0.380)
<i>T</i> - 1	0.465 (0.398)
<i>T</i> + 0	1.741*** (0.657)
<i>T</i> + 1	1.922*** (0.604)
<i>T</i> + 2	0.948** (0.398)
<i>T</i> + 3	0.436 (0.408)
<i>tenure</i>	-0.041 (0.076)
<i>project</i>	0.003** (0.001)
<i>sponsored</i>	0.089 (0.291)
<i>public</i>	2.167*** (0.291)
<i>release</i>	0.474*** (0.130)
Constant	2.386 (4.797)
Individual FE	Yes
Time FE	Yes
Observations	5,851,040

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Heteroskedasticity-robust standard errors in parentheses.

We conduct a placebo test to examine whether our results are driven solely by chance. Following Bertrand et al. (2004), we randomly generate placebo users' sponsorships, and treated users are chosen at random. Because these users' sponsorships are fictitious, a significant "effect" of providing sponsorship at the 5% significance level should be found approximately 5% of the time. If, in the placebo test, a significant effect at the 5% significance level is found at a value much larger than 5% of the time, our estimated treatment effects can be driven by the placebo effect. This means that we do not have significant evidence of the positive effect of sponsoring others on project popularity. By contrast, if we find a significant effect at the 5% level at a value of around 5%, the test provides evidence that the positive effect of sponsoring others is not driven by the placebo effect.

We randomly select 12,520 users from our sample, which is equal to the number of real sponsors, and designate them as treated users. We then estimate Equation (1) using these placebo sponsorships, generating an estimate of the effect of providing sponsorship and a standard error for this estimate. We repeat this process 500 times. Since these users' sponsorships are fictitious, we expect to reject the null hypothesis of no effect approximately 5% of the time (25 times). Our results from the 500 runs show that the null hypothesis is rejected 23 times, i.e., 4.6% of the time. This finding suggests that our estimated effects are unlikely to be driven by the placebo effect.

DID Combined with PSM

We also utilize a combination of the DID model and the PSM method to mitigate any bias between the control and treatment groups. The underlying principle is to form matched pairs of treated and nontreated

units with similar observed characteristics and nearly identical propensities of being treated. Then, we can use the matched sample to run the DID regression.

PSM can be implemented in two different ways: static and dynamic. In both methods, the treated users are those who have sponsored others. In the static implementation, propensity scores of all the sample sponsors are tabulated. These treated users are matched with non-sponsors who resemble them most closely in terms of their overall propensity scores based on the data until May 23, 2019. Under static matching, the matched pool of control users remains the same for all months during the study period. In the dynamic implementation, unique propensity scores are calculated for sponsors each month, enabling them to be matched with different non-sponsors over time.

In calculating the propensity scores, we use a set of pre-treatment covariates that are likely to influence a user's sponsoring behavior and OSS project popularity (Borges & Tulio Valente, 2018; Krishnamurthy & Tripathi, 2009; Weng & Soh, 2023): the number of months since the user joined GitHub (*tenure*), whether the user is affiliated with a *company* according to his or her profile page, the total number of OSS *projects* that the user created, the total number of sponsorships that the user received, whether the user leaves his or her *contact* information on the profile page, the number of *stars* that the user received during the past year, the number of repositories that the user turned from private to *public* during the past year, and the number of *releases* in the user's OSS projects during the past year. We use the same set of covariates for both matching procedures, except for the total number of sponsorships received. This covariate is omitted in static matching since no user could have received sponsorship before May 23, 2019, resulting in a value of zero for all users in the static method.

Table 3. Differences in Mean Before and After One-to-Two Static Matching

Variable	Before matching				After matching			
	Mean treated	Mean control	<i>t</i> -statistics	<i>p</i> -value	Mean treated	Mean control	<i>t</i> -statistics	<i>p</i> -value
<i>tenure</i>	78.444	49.631	108.25***	0.000	78.444	78.785	-0.90	0.369
<i>company</i>	0.523	0.234	70.43***	0.000	0.523	0.529	-0.95	0.343
<i>project</i>	39.967	15.773	32.68***	0.000	39.967	41.968	-0.73	0.464
<i>contact</i>	0.804	0.443	78.38***	0.000	0.804	0.810	-1.11	0.269
<i>star</i>	96.921	9.669	23.18***	0.000	96.921	110.99	-1.11	0.267
<i>public</i>	0.475	0.185	32.91***	0.000	0.475	0.482	-0.32	0.753
<i>release</i>	2.346	0.386	28.89***	0.000	2.346	2.051	1.38	0.168

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

Table 4. Differences in Mean Before and After One-to-Three Static Matching

Variable	Before matching				After matching			
	Mean treated	Mean control	<i>t</i> -statistics	<i>p</i> -value	Mean treated	Mean control	<i>t</i> -statistics	<i>p</i> -value
<i>tenure</i>	78.444	49.631	108.25***	0.000	78.444	78.621	-0.47	0.641
<i>company</i>	0.523	0.234	70.43***	0.000	0.523	0.529	-1.05	0.295
<i>project</i>	39.967	15.773	32.68***	0.000	39.967	39.626	0.15	0.882
<i>contact</i>	0.804	0.443	78.38***	0.000	0.804	0.811	-1.38	0.168
<i>star</i>	96.921	9.669	23.18***	0.000	96.921	105.78	-0.72	0.471
<i>public</i>	0.475	0.185	32.91***	0.000	0.475	0.478	-0.12	0.906
<i>release</i>	2.346	0.386	28.89***	0.000	2.346	2.123	1.00	0.317

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

In both the static and dynamic procedures, we employ two matching algorithms (i.e., one-to-two and one-to-three nearest neighbor matching with replacement) to assess the robustness of the results across various matched samples. To assess the quality of static matching, we conducted *t*-tests of equality of means before and after the matching process to ensure that our static matching method adequately balanced the observed characteristics between the treatment and control groups. We present the results of one-to-two and one-to-three matching in Tables 3 and 4, respectively. These two tables show that there is a significant covariate

imbalance between the treatment and control groups before matching. However, after matching, the differences in means are no longer statistically significant, indicating that both matching algorithms effectively reduce the bias associated with the observed characteristics. It is worth mentioning that after matching, the number of stars received by the control group during the year prior to the introduction of the peer-to-peer sponsorship feature is not statistically different from that of the treated group, suggesting that our matching is successful. The *t*-test results of dynamic matching are similar to those of static matching and are not presented here, as each dynamic matching algorithm was run 40 times (for each month in our study period following the introduction of the peer-to-peer sponsorship feature).

Next, we re-estimate Equations (1)-(3) using the new samples created by the four matching algorithms, and the results are presented in Tables 5 and 6. We find that users who sponsor others on GitHub receive around 2 more stars for their projects than non-sponsors. Moreover, this positive impact of providing sponsorship on project popularity is larger for individuals with shorter tenure in the community and those without announced affiliation. All these results are consistent with those in Table 1.

Table 5. Results Based on Static PSM

	Dependent variable: <i>star</i>					
	One-to-two static PSM			One-to-three static PSM		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>sponsor</i>	2.526*** (0.427)	5.828*** (1.365)	3.476*** (0.603)	2.493*** (0.425)	5.887*** (1.365)	3.439*** (0.598)
<i>sponsor × tenure</i>		-0.030** (0.012)			-0.031*** (0.012)	
<i>sponsor × company</i>			-1.801** (0.830)			-1.792** (0.830)
<i>tenure</i>	0.370 (0.286)	0.375 (0.286)	0.369 (0.286)	0.469* (0.245)	0.472* (0.245)	0.468* (0.245)
<i>project</i>	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
<i>sponsored</i>	0.159 (0.283)	0.160 (0.283)	0.158 (0.283)	0.174 (0.273)	0.175 (0.273)	0.173 (0.273)
<i>public</i>	3.285*** (0.418)	3.287*** (0.418)	3.285*** (0.418)	3.166*** (0.345)	3.168*** (0.345)	3.166*** (0.345)
<i>release</i>	0.666*** (0.164)	0.666*** (0.164)	0.667*** (0.164)	0.637*** (0.133)	0.637*** (0.133)	0.638*** (0.133)
Constant	-17.608 (20.464)	-17.958 (20.452)	-17.592 (20.465)	-24.856 (17.471)	-25.121 (17.463)	-24.844 (17.472)
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,953,120	1,953,120	1,953,120	2,604,160	2,604,160	2,604,160

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Heteroskedasticity-robust standard errors in parentheses.

Table 6. Results Based on Dynamic PSM

	Dependent variable: <i>star</i>					
	One-to-two dynamic PSM			One-to-three dynamic PSM		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>sponsor</i>	1.718*** (0.469)	5.399*** (1.363)	2.644*** (0.625)	1.849*** (0.457)	5.539*** (1.362)	2.771*** (0.615)
<i>sponsor × tenure</i>		-0.034*** (0.012)			-0.034*** (0.012)	
<i>sponsor × company</i>			-1.756** (0.829)			-1.747** (0.829)
<i>tenure</i>	0.539*	0.544*	0.539*	0.323	0.327	0.323

	(0.317)	(0.317)	(0.317)	(0.284)	(0.283)	(0.284)
<i>project</i>	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.001)	0.002*** (0.000)	0.002*** (0.000)	0.002*** (0.000)
<i>sponsored</i>	0.470* (0.272)	0.471* (0.272)	0.470* (0.272)	0.504* (0.262)	0.505* (0.262)	0.504* (0.262)
<i>public</i>	4.518*** (0.500)	4.520*** (0.500)	4.517*** (0.500)	4.073*** (0.395)	4.074*** (0.395)	4.072*** (0.395)
<i>release</i>	0.675*** (0.142)	0.675*** (0.142)	0.676*** (0.142)	0.661*** (0.125)	0.661*** (0.124)	0.661*** (0.124)
Constant	-32.941 (22.210)	-33.285 (22.194)	-32.925 (22.211)	-17.544 (19.899)	-17.803 (19.890)	-17.532 (19.900)
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,952,652	1,952,652	1,952,652	2,603,536	2,603,536	2,603,536

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Heteroskedasticity-robust standard errors in parentheses.

Robustness Checks

In this section, we enhance the robustness of our findings by exploring additional tests. Specifically, we address the multicollinearity issue, apply log-transformation and random effects models, control the lagged dependent variable, and estimate models with no control variables.

Multicollinearity

We address the issue of multicollinearity in Equations (1)-(3) by examining the variance inflation factors (VIFs). The VIF quantifies the degree of multicollinearity in a regression. When the independent variables are correlated, the estimated standard errors of the fitted coefficients are inflated, which can be used to check for the presence of multicollinearity (Chatterjee & Hadi, 2012). Specifically, a VIF of 1 for an independent variable indicates no linear relationship with other independent variables. Any deviation from this value suggests a departure from orthogonality and a tendency toward collinearity. As suggested by Chatterjee and Hadi (2012), “values of variance inflation factors greater than 10 is often taken as a signal that the data have collinearity problems.” In our analysis, we examine the VIFs to evaluate possible multicollinearity among model covariates. Our results show that the VIFs associated with each variable are not greater than 1.12 (see Table 7), which is well below the conventionally accepted threshold of 10, indicating that multicollinearity is not an issue in our models and does not significantly affect our findings.

Table 7. Collinearity Diagnostics	
	VIF
<i>sponsor</i>	1.08
<i>tenure</i>	1.12
<i>company</i>	1.05
<i>project</i>	1.02
<i>sponsored</i>	1.00
<i>public</i>	1.00
<i>release</i>	1.00
Mean VIF	1.04

Log-Transformation Models

To mitigate the skewness of our dependent variable, $star_{it}$, we employ log-transformation models as a robustness check. To account for instances where a user receives zero star for his or her projects, we add 0.01 to the number of stars received by a user before taking the natural logarithm. We then use this transformed dependent variable to re-estimate Equations (1)-(3). The estimation results, presented in Table 8, are consistent with our previous findings: providing peer-to-peer sponsorship has a positive impact on

project popularity, and this impact is larger for shorter-tenured individuals and individuals without announced affiliation.

Table 8. Impacts of Providing Sponsorship on Project Popularity: Log-Transformation Models			
	Dependent variable: $\ln(star + 0.01)$		
	(1)	(2)	(3)
<i>sponsor</i>	0.209*** (0.013)	0.881*** (0.045)	0.243*** (0.017)
<i>sponsor × tenure</i>		-0.006*** (0.000)	
<i>sponsor × company</i>			-0.064** (0.025)
<i>tenure</i>	0.001 (0.003)	0.001 (0.003)	0.001 (0.003)
<i>project</i>	0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
<i>sponsored</i>	-0.002* (0.001)	-0.002* (0.001)	-0.002* (0.001)
<i>public</i>	0.296*** (0.009)	0.296*** (0.009)	0.296*** (0.009)
<i>release</i>	0.027*** (0.004)	0.027*** (0.004)	0.027*** (0.004)
Constant	-4.023*** (0.138)	-4.038*** (0.138)	-4.023*** (0.138)
Individual FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Observations	5,851,040	5,851,040	5,851,040

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Heteroskedasticity-robust standard errors in parentheses.

Random Effects Models

While the results from the Hausman test indicate that the fixed effects model is more appropriate than the random effects model, we also undertake an analysis using individual random effects instead of fixed effects. This alternative approach serves to control for any unobserved time-invariant individual-level effects and facilitates between-individual comparisons (Zhang, 2020). As shown in Table 9, the results of the three random effects models are consistent with those in Table 1.

Table 9. Impacts of Providing Sponsorship on Project Popularity: Log-Transformation Models			
	Dependent variable: <i>star</i>		
	(1)	(2)	(3)
<i>sponsor</i>	1.222*** (0.395)	5.006*** (1.342)	2.141*** (0.588)
<i>sponsor × tenure</i>		-0.035*** (0.012)	
<i>sponsor × company</i>			-1.742** (0.827)
<i>tenure</i>	0.055*** (0.006)	0.056*** (0.006)	0.055*** (0.006)
<i>project</i>	0.003* (0.002)	0.003* (0.002)	0.003* (0.002)
<i>sponsored</i>	0.052	0.052	0.052

	(0.122)	(0.122)	(0.122)
<i>public</i>	2.202*** (0.292)	2.203*** (0.292)	2.202*** (0.292)
<i>release</i>	0.503*** (0.140)	0.503*** (0.140)	0.503*** (0.140)
Constant	-1.296 (1.052)	-1.354 (1.051)	-1.301 (1.052)
Individual RE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Observations	5,851,040	5,851,040	5,851,040

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Heteroskedasticity-robust standard errors in parentheses.

Lagged Dependent Variable

To alleviate the concern that users receive more stars for their projects due to their pre-existing popularity, we include the lagged dependent variable as a control variable in Equations (1)-(3) (Zhang, 2020). Our estimation results, presented in Table 10, remain consistent with our earlier findings.

Table 10. Impacts of Providing Sponsorship on Project Popularity: Including Lagged Dependent Variable			
	Dependent variable: <i>star</i>		
	(1)	(2)	(3)
<i>sponsor</i>	0.659*** (0.252)	3.056*** (0.829)	1.247*** (0.357)
<i>sponsor × tenure</i>		-0.022*** (0.007)	
<i>sponsor × company</i>			-1.116** (0.523)
<i>tenure</i>	-0.045 (0.053)	-0.044 (0.053)	-0.045 (0.053)
<i>project</i>	0.002** (0.001)	0.002** (0.001)	0.002** (0.001)
<i>sponsored</i>	0.046 (0.180)	0.047 (0.180)	0.046 (0.180)
<i>public</i>	2.110*** (0.284)	2.110*** (0.284)	2.110*** (0.284)
<i>release</i>	0.358*** (0.094)	0.358*** (0.094)	0.359*** (0.094)
<i>star (t - 1)</i>	0.361*** (0.035)	0.361*** (0.035)	0.361*** (0.035)
Constant	4.670 (5.621)	4.578 (5.619)	4.672 (5.621)
Individual FE	Yes	Yes	Yes
Time FE	Yes	Yes	Yes
Observations	5,738,520	5,738,520	5,738,520

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Heteroskedasticity-robust standard errors in parentheses.

Endogenous Control Variables

To mitigate the potential influence of endogenous control variables, we compare the results from models without controls with those in Table 1. The results for comparison are reported in Table 11. We find that the coefficients of the focal variables remain consistent, regardless of the inclusion or omission of controls. This finding suggests that the concern of endogenous controls might be alleviated to some extent. When the coefficient of the focal variable remains stable after incorporating the observed controls, it is generally

interpreted as an indication that concerns about omitted variable bias are limited (Chiappori et al., 2012; Khurana et al., 2019).

Table 11. Impacts of Providing Sponsorship on Project Popularity: Control Variables vs. No Control Variables						
	Dependent variable: <i>star</i>					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>sponsor</i>	1.380*** (0.416)	1.089*** (0.407)	5.090*** (1.351)	5.100*** (1.369)	2.254*** (0.594)	2.027*** (0.584)
<i>sponsor × tenure</i>			-0.035*** (0.012)	-0.037*** (0.012)		
<i>sponsor × company</i>					-1.683** (0.832)	-1.778** (0.832)
<i>tenure</i>		-0.041 (0.076)	0.005** (0.001)	-0.039 (0.076)		-0.041 (0.076)
<i>project</i>		0.003** (0.001)		0.003** (0.001)		0.003** (0.001)
<i>sponsored</i>		0.088 (0.291)		0.090 (0.291)		0.088 (0.291)
<i>public</i>		2.170*** (0.291)		2.171*** (0.291)		2.170*** (0.291)
<i>release</i>		0.473*** (0.130)		0.473*** (0.130)		0.474*** (0.130)
<i>company</i>					2.647*** (0.225)	
Constant	1.761*** (0.095)	2.394 (4.795)	1.425*** (0.113)	2.305 (4.794)	1.058*** (0.108)	2.399 (4.796)
Individual FE	No	Yes	No	Yes	No	Yes
Time FE	No	Yes	No	Yes	No	Yes
Observations	5,851,040	5,851,040	5,851,040	5,851,040	5,851,040	5,851,040

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$. Heteroskedasticity-robust standard errors in parentheses.

It is important to note that even in cases where control variables exhibit endogeneity (correlation with the error term), their inclusion may not necessarily influence the estimates of the focal variables. Stock (2010) emphasizes that when certain conditions are met, such as the conditional mean independence assumption, coefficients of the focal variables can remain unbiased and consistent in the presence of endogenous control variables, but the same attribute of unbiasedness and consistency may not apply to coefficients of the endogenous control variables.

Conclusion

The emergence of peer-to-peer sponsorship in online communities has rendered it a significant and compelling topic for scholarly investigation. While the impacts of peer-to-peer sponsorship on the sponsees have been extensively explored (Medappa et al., 2023; Shimada et al., 2022; J. Wang et al., 2022), the impacts on sponsors, particularly concerning their potential gains, remain largely unknown. To fill this gap, we leverage the introduction of GitHub Sponsors to investigate the impact of providing sponsorship on project popularity and the moderating effects of community tenure and affiliation disclosure. Our findings indicate that sponsoring others enhances the popularity of an individual's OSS projects. Moreover, individuals with shorter tenure and those without announced affiliations within the community experience a greater increase in project popularity than those with longer tenure and affiliated affiliations after sponsoring others.

Although our study provides valuable insights into the impact of sponsoring others on project popularity, there are several limitations that future research could address. First, we did not investigate the impact of sponsorship amount due to the unavailability of such data on GitHub. Future research with available data

could overcome this limitation by exploring the impact of sponsorship amount on project popularity. Second, our study focuses on the impact on project popularity. Future research could investigate the impact of providing sponsorship on other aspects of sponsors, such as their offline behavior and how it relates to their online behavior. Third, to increase the generalizability of our findings, future research could extend our study to other online communities beyond GitHub. This would help to determine the extent to which our results apply to other contexts. Last, future research may explore whether different factors, like team belonging, can lead to varying effects.

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